

João Vitor Bermal Santaniello

Software Engineer · Full-Stack, AI-Augmented Delivery and Geospatial Data

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SUMMARY

Software engineer with six years of production experience in water utilities, energy, healthcare and telecom. Main developer of EyeConnect, a teleophthalmology platform in production, and of the ISPDrive rewrite. For Aegea I cut a geospatial workflow from 3+ hours to about 20 minutes. AI-native with Claude Code and MCP. MBA (USP/ESALQ, thesis 9/10), fluent English.

EXPERIENCE

Software Developer, Bizpoke Soluções em Software · São José dos Campos, SP Dec 2022 to Present

- Cut a recurring geospatial workflow at Aegea from more than 3 hours to about 20 minutes with a reusable Python and GeoPandas tool, and delivered census tract indicators for 16 states for concession studies.
- Co-designed the CITIUS/Regenera landfill biogas data model and kept production ArcGIS dashboards running through the Enterprise 11.5 migration.
- Main developer of EyeConnect (teleophthalmology, in production): Raspberry Pi DICOM gateway and OCR exam intake with AWS Textract.
- Rewrote ISPDrive, a multi-tenant storage SaaS for internet providers, in FastAPI, MongoDB and Next.js 14.
- Authored nine shared Claude Code skills and MCP connections (ClickUp, the bizpoke MCP server) used across 13 Aegea projects.
- Led front-end and mobile delivery for DOMO (Equatorial Energia): offline map editing in React Native plus Next.js, TypeScript and Redux screens.

AI Coding Benchmark Contributor (freelance), Alignerr · Remote May 2025 to Present

- Wrote Python problems, reference solutions and test suites for LiveCodeBench and BigCodeBench, each designed to break at least 2 of 4 frontier code models.

Software Development Intern, Bizpoke Soluções em Software · São José dos Campos, SP Apr 2021 to Dec 2022

- Shipped React, Node.js, Python and C# features and produced spatial data with ArcGIS Pro and Enterprise alongside offshore teams.

SELECTED PROJECTS

Triage Desk, AI support triage for D2C brands · 2026 triage.joaosantaniello.com

Live demo: Claude agent with sub-agents in a Supabase Edge Function, n8n workflows, RLS with pgTAP tests, Next.js on Vercel.

Code: github.com/joao-bermal/triage-desk

EDUCATION

MBA in Software Engineering, Universidade de São Paulo (USP/ESALQ) 2024 to 2026

Thesis graded 9/10: deep learning pipeline for retinal images (vessel Dice 0.79 on held-out data, optic disc error cut from 358 to 11 px).

Associate degree in Systems Analysis and Development, Universidade Paulista (UNIP) 2021 to 2022

Technical Program in Computing, with high school, Colégio Técnico Antônio Teixeira Fernandes 2018 to 2020

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Bash

Data and GIS: GeoPandas, ArcGIS Pro, ArcGIS Enterprise 11.5, ArcGIS Dashboards, Experience Builder, ArcGIS JS API, FME, PostgreSQL, enterprise geodatabases, ETL and data modeling

Back end: FastAPI, Node.js, Meteor, REST APIs, JWT, OAuth, WebSockets, MongoDB, PostgreSQL, Supabase, DICOM

Front end and mobile: React, Next.js, Redux Toolkit, Tailwind CSS, MUI, React Native (Expo), Figma

Cloud and DevOps: Docker, NGINX, Linux, AWS (S3, Textract, EC2), DigitalOcean, Vercel, n8n, basic Kubernetes

AI and ML: Claude Code (skills, MCP, sub-agents), Claude API, PyTorch, OpenCV, OCR (Textract), LLM evaluation

LANGUAGES

Portuguese (native) · English (fluent, daily professional use)